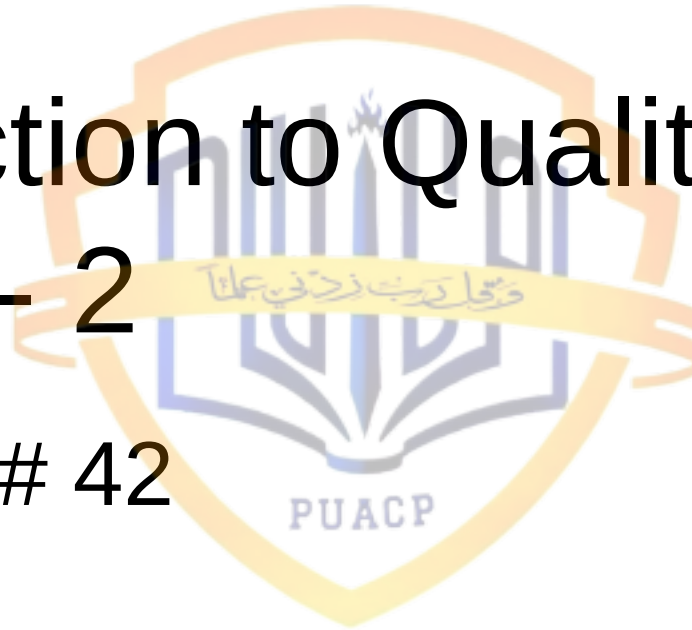


Introduction to Quality Metrics - 2

Lecture # 42



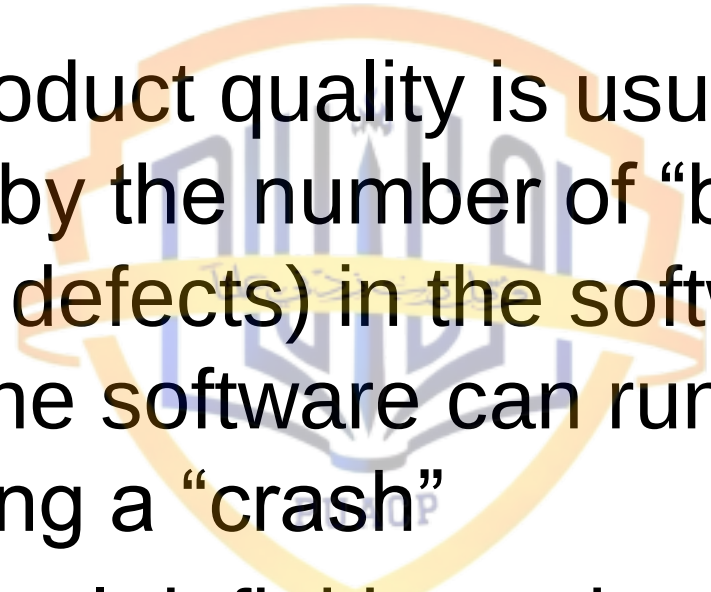
Seven Commonly Tracked Measures

- Number of defects
- Work effort
- Schedule
- Number of changes to the requirements
- Size
- Documentation defects
- Complexity



Additional Software Quality Metrics

- Mean time to failure (MTTF)
- Defect density
- Defects by severity
- Customer problems
- Customer satisfaction

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- Intrinsic product quality is usually measured by the number of “bugs” (functional defects) in the software or by how long the software can run before encountering a “crash”
 - In operational definitions, the two metrics are defect density (rate) and mean time to failure (MTTF)

Mean Time To Failure - 1

- The mean time to failure metric measures the time between failures
- It is often used with safety-critical systems such as the airline traffic control systems, avionics, and weapons
- For example, the air traffic control system cannot be unavailable for more than three seconds a year

Mean Time To Failure - 2

- A failure occurs when a functional unit of a software-related system can no longer perform its required function or cannot perform it within specified limits
- It requires that the profile of a system is available to know the tolerance
- Mean time between failures (MTBF)

Defect Density - 1

- The defect density measures the number of defects discovered per some unit of software size (lines of code, function points)
- The defect density metric is used in many commercial software systems
- The defect rate of a product or the expected number of defects over a certain time period is important for cost and resource estimates of the maintenance phase of the software life cycle

Defect Density - 2

- The defect density metric is used to measure the number of defects discovered per some unit of product size (KLOC or function points)
- For example, if your product has 10 KLOC and 127 defects were discovered during a test cycle, the defect density would be 0.0127 defects per KLOC

Defect Density - 3

- The defect density metric can be applied during any test period; however, only the value calculated during test phases that follow system integration can be used to make predictions about the rate at which defects will be discovered by customers

Defects by Severity - 1

- The defects by severity metric is a simple count of the number of unresolved defects listed by severity
- Typically, this metric is measured at some regular interval and plotted to determine whether or not a trend exists

Defects by Severity - 2

- Ideally, a trend exists, showing progress toward the acceptable values for each severity
- Movement away from those values should raise a flag that the project is at risk of failing to satisfy the conditions of the metric

Customer Problems - 1

- This metric is a simple count of the number of new (non-duplicate) problems reported by customers over some time interval
- When measured at regular intervals and plotted, the data can be used to identify a trend. Although, a trend may be apparent, it is more useful to determine the reasons behind the trend

Customer Problems - 2

- If, for example, the number of customer-reported problems increases over time, is it because more end users are using the product?
- If you measure the number of customers who use the product at the same intervals that you measure customer-reported problems, you might identify a cause-effect or correlation between the metric and number of end users

Customer Problems - 3

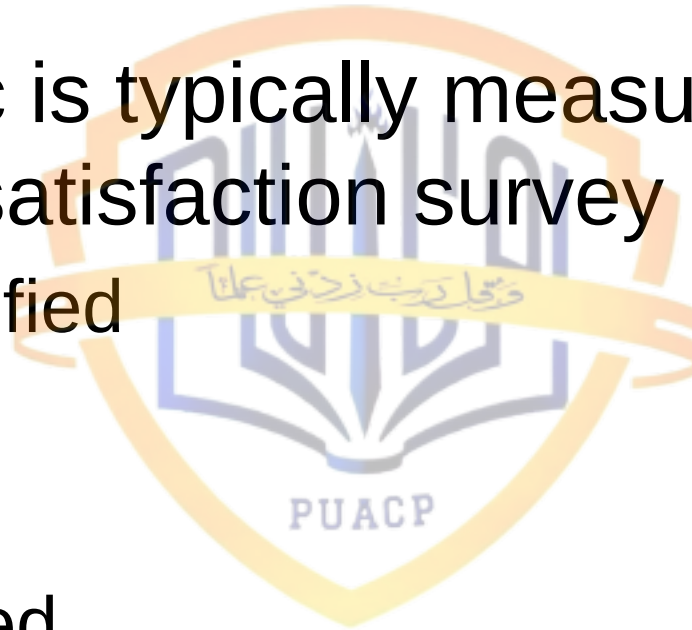
- For example, if you determine that as the number of end users of the system increases the number of customer-reported problems increases, a relationship may exist between the two that suggests that you may have a serious scalability flaw in your product

Customer Problems - 4

- On the other hand, is the increase related to greater demands placed on the system by end users as their experience with the product matures? With the help of profiling features, you can determine the load on the product

Customer Satisfaction - 1

- This metric is typically measured through a customer satisfaction survey
 - ☐ Very satisfied
 - ☐ Satisfied
 - ☐ Neutral
 - ☐ Dissatisfied
 - ☐ Very dissatisfied



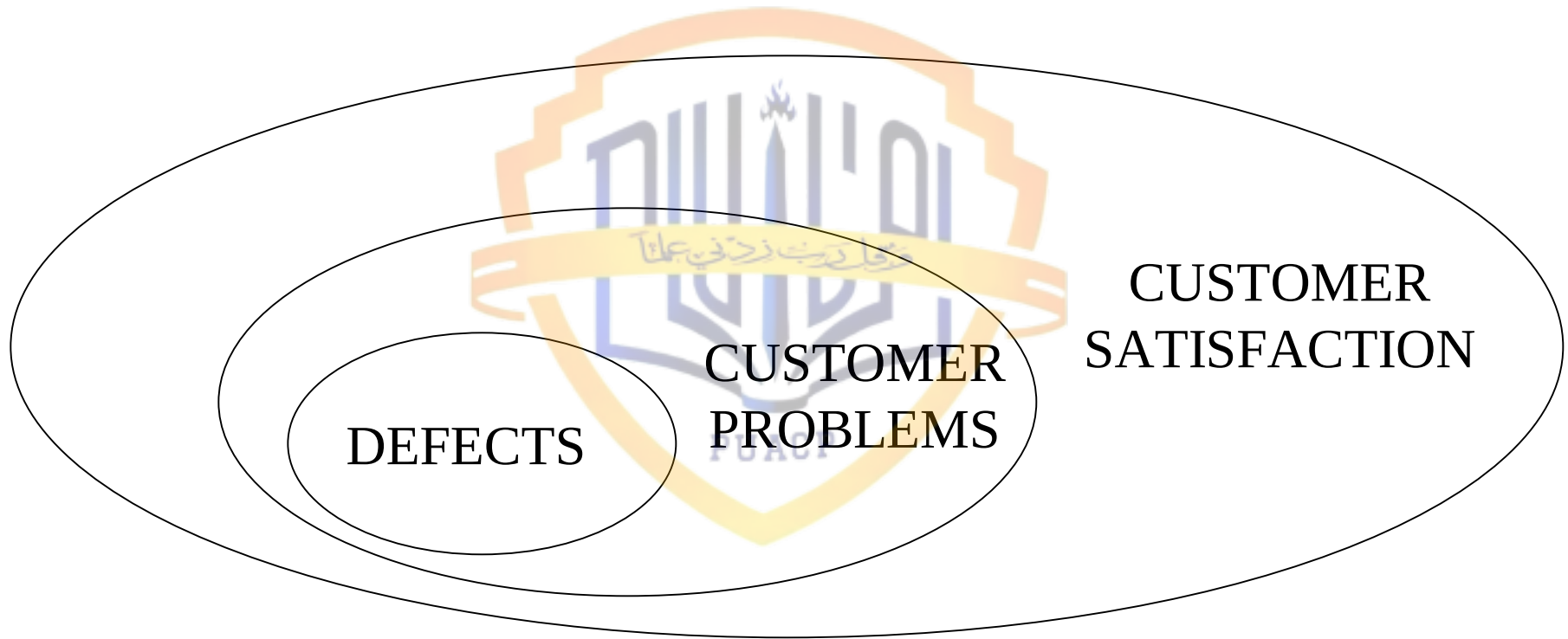
Customer Satisfaction - 2

- Percent of completely satisfied customers
- Percent of satisfied customers
 - Satisfied and completely satisfied
- Percent of dissatisfied customers
 - Dissatisfied and completely dissatisfied
- Percent of non-satisfied customers
 - Neutral, dissatisfied and completely dissatisfied



Scopes of Three Quality Metrics

Scopes of Three Quality Metrics

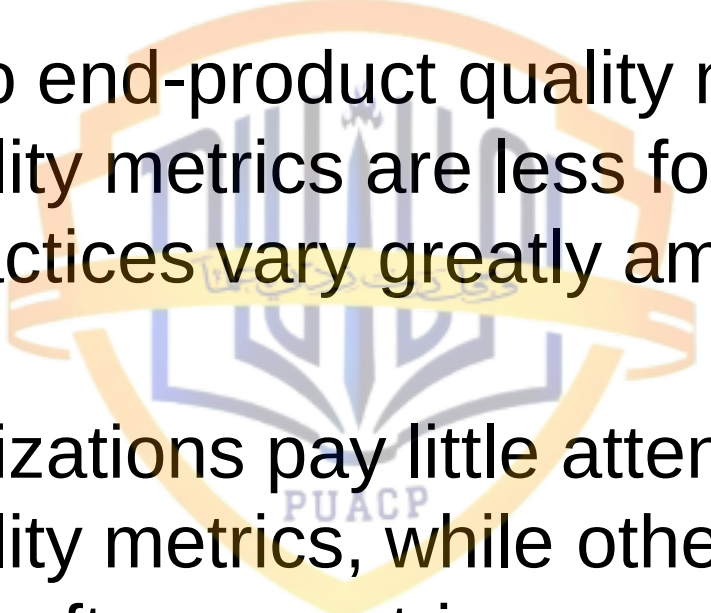


Process Quality Metrics



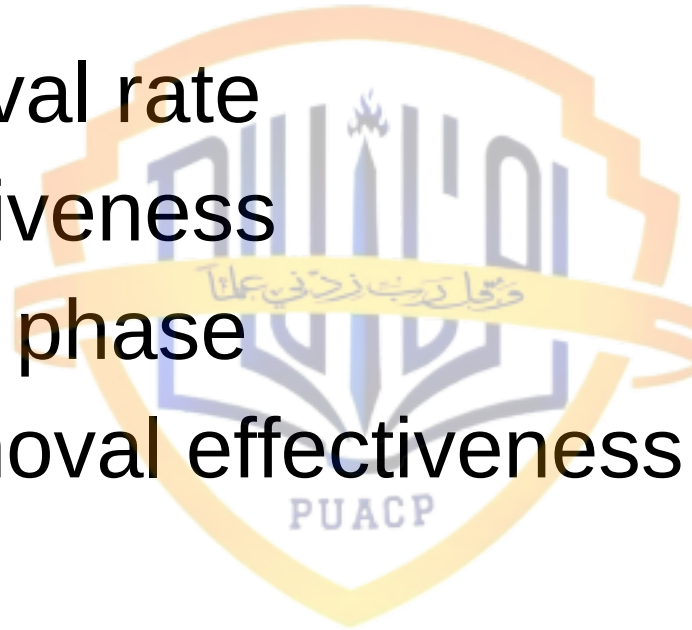
Process Metrics

- Process metrics are those that can be used for improving the software development and maintenance process
- Examples include the effectiveness of defect removal during development, the pattern of testing defect arrival, and the response time of the fix process

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- Compared to end-product quality metrics, process quality metrics are less formally defined, and their practices vary greatly among software developers
 - Some organizations pay little attention to process quality metrics, while others have well-established software metrics programs that cover various parameters in each phase of the development cycle

Process Metrics

- Defect arrival rate
- Test effectiveness
- Defects by phase
- Defect removal effectiveness



Defect Arrival Rate – 1

- It is the number of defects found during testing measured at regular intervals over some period of time
- Rather than a single value, a set of values is associated with this metric
- When plotted on a graph, the data may rise, indicating a positive defect arrival rate; it may stay flat, indicating a constant defect arrival rate; or decrease, indicating a negative defect arrival rate

Defect Arrival Rate – 2

- Interpretation of the results of this metric can be very difficult
- Intuitively, one might interpret a negative defect arrival rate to indicate that the product is improving since the number of new defects found is declining over time
- To validate this interpretation, you must eliminate certain possible causes for the decline

Defect Arrival Rate – 3

- For example, it could be that test effectiveness is declining over time. In other words, the tests may only be effective at uncovering certain types of problems. Once those problems have been found, the tests are no longer effective

Defect Arrival Rate – 4

- Another possibility is that the test organization is understaffed and consequently is unable to adequately test the product between measurement intervals. They focus their efforts during the first interval on performing stress tests that expose many problems, followed by executing system tests during the next interval where fewer problems are uncovered

Test Effectiveness - 1

- To measure test effectiveness, take the number of defects found by formal tests and divide by the total number of formal tests
- $TE = D_n / T_n$

Test Effectiveness - 2

- When calculated at regular intervals and plotted, test effectiveness can be observed over some period of time
- If the graph rises over time, test effectiveness may be improving. On the other hand, if the graph is falling over time, test effectiveness may be waning

Defects by Phase - 1

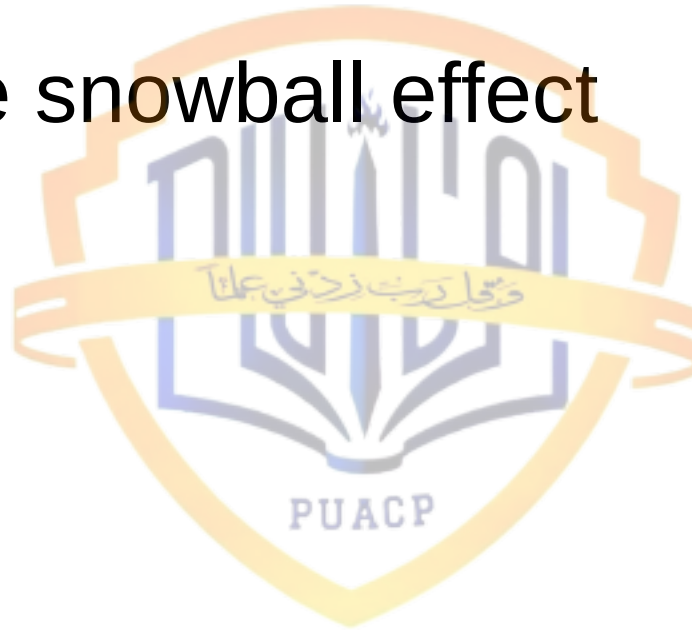
- It is much less expensive in terms of resources and reputation to eliminate defects early instead of fix late
- The defects by phase metric is a variation of the defect arrival rate metric
- At the conclusion of each discrete phase of the development process, a count of the new defects is taken and plotted to observe a trend

Defects by Phase - 2

- If the graph appears to be rising, you might infer that the methods used for defect detection and removal during the earlier phases are not effective since the rate at which new defects are being discovered is increasing
- On the other hand, if the graph appears to be falling, you might conclude that early defect detection and removal is effective

Defects by Phase - 3

- Explain the snowball effect



Defect Removal Effectiveness - 1

- Defect removal effectiveness can be calculated by dividing the number of defects removed prior to release by the sum of defects removed prior to release and the total number of defects that remain in the product at release. When multiplied by 100, this value can be expressed as a percentage:
- $DRE = D_r / (D_r + D_t)$

Defect Removal Effectiveness - 2

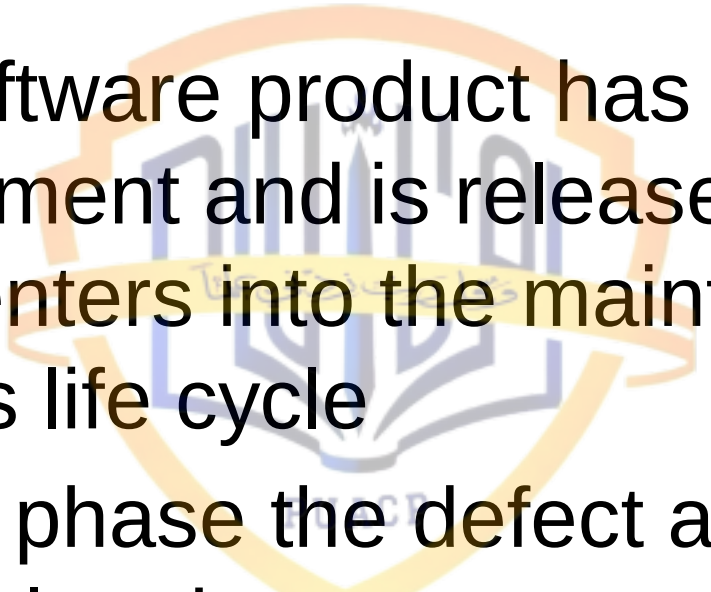
- (say more on this)





Metrics for Software Maintenance Process



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- When a software product has completed its development and is released to the market, it enters into the maintenance phase of its life cycle
 - During this phase the defect arrivals by time interval and customer problem calls (which may or may not be defects) by time interval rate are the *de facto* metrics

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- However, the number of problem arrivals is largely determined by the development process before the maintenance phase
 - Not much can be done to alter the quality of the product during this phase.
- Therefore, the *de facto* metrics, while important, do not reflect the quality of software maintenance

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- What can be done during maintenance phase is to fix the defects as soon as possible and with excellent fix quality
 - Such actions, although still not able to improve the defect rate of the product, can improve customer satisfaction to a large extent

Metrics in Software Maintenance

- Defect backlog
- Backlog management index
- Fix response time
- Percent delinquent fixes
- Defective fixes



Defect Backlog - 1

- The defect backlog metric is a count of the number of defects in the product following its release that require a repair
- It is usually measured at regular intervals of time and plotted for trend analysis

Defect Backlog - 2

- By itself, this metric provides very little useful information
- For example, what does a defect backlog count of 128 tell you? Can you predict the impact of those defects on customers? Can you estimate the time it would take to repair those defects? Can you recommend changes to improve the development process?

Defect Backlog - 3

- A more useful way to represent the defect backlog is by defect severity
- By calculating the defect backlog by severity level, you can begin to draw useful conclusions from your measurements

Backlog Management Index - 1

- As the backlog is worked, new problems arrive that impact the net result of your team's efforts to reduce the backlog
- If the number of new defects exceeds the number of defects closed over some period of time, your team is losing ground to the backlog. If, on the other hand, your team closes problems faster than new ones are opened, they are gaining ground

Backlog Management Index - 2

- The backlog management index (BMI) is calculated by dividing the number of defects closed during some period of time by the number of new defects that arrived during that same period of time
- $BMI = D_c / D_n$

Backlog Management Index - 3

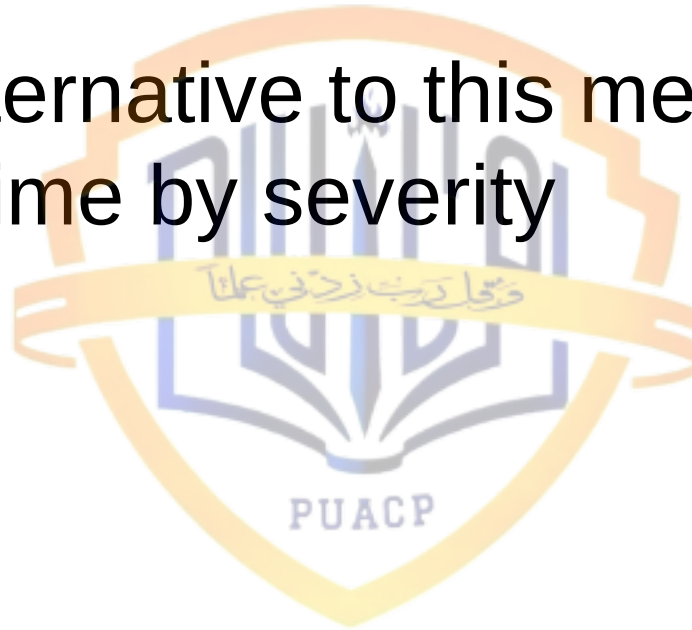
- If the result is greater than 1, your team is gaining ground; otherwise it is losing
- When measurements are taken at regular intervals and plotted, a trend can be observed indicating the rate at which the backlog is growing or shrinking

Fix Response Time - 1

- The fix response time metric is determined by calculating the average time it takes your team to fix a defect
- It can be measured several different ways
- In some cases, it is the elapsed time between the discovery of the defect and the development of an unverified fix
- In other cases, it is the elapsed time between the discovery and the development of verified fix

Fix Response Time - 2

- A better alternative to this metric is fix response time by severity



Percent Delinquent Fixes - 1

- A fix is delinquent if exceeds your fix response time criteria. In other words, if you have established a maximum fix response time of 48 hours; then fix response times that exceed 48 hours are considered delinquent
- To calculate the percent delinquent fixes; divide the number of delinquent fixes by the number of non-delinquent fixes and multiply by 100
- $PDF = (F_d / F_n) * 100$

Percent Delinquent Fixes - 2

- This metric is also measured better by severity since the consequences of having a high delinquent fixes for severe defects is typically much greater than for less severe or minor defects

Defective Fixes - 1

- A defect for which a fix has been prepared that later turns out to be defective or worse, creates one or more additional problems, is called a defective fix
- The defective fixes metric is a count of the number of such fixes

Defective Fixes - 2

- To accurately measure the number of defective fixes, your organization must not only keep track of defects that have been closed and then reopened but must also keep track of new defects that were caused by a defect fix
- This metric is also known as fix quality

Summary



References

- Customer-Oriented Software Quality Assurance by Frank P. Ginac (Chapter 3)
- Metrics and Models in Software Quality Engineering by Stephan H. Kan (Chapter 4)